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Fine-Tuning BERT for Multi-Label Sentiment Analysis in Unbalanced Code-Switching Text

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ABSTRACT Previous research on sentiment analysis mainly focuses on binary or ternary sentiment analysis in monolingual texts. However, in today's social media such as micro-blogs, emotions are often expressed in bilingual or multilingual text called code-switching text, and people's emotions are complex, including happiness, sadness, angry, afraid, surprise, etc. Different emotions may exist together, and the proportion of each emotion in the code-switching text is often unbalanced. Inspired by the recently proposed BERT model, we investigate how to fine-tune BERT for multi-label sentiment analysis in code-switching text in this paper. Our investigation includes the selection of pre-trained models and the fine-tuning methods of BERT on this task. To deal with the problem of the unbalanced distribution of emotions, a method based on data augmentation, undersampling and ensemble learning is proposed to get balanced samples and train different multi-label BERT classifiers. Our model combines the prediction of each classifier to get the final outputs. The experiment on the dataset of NLPCC 2018 shared task 1 shows the effectiveness of our model for the unbalanced code-switching text. The F1-Score of our model is higher than many previous models.

INDEX TERMS Text analysis, classification algorithms, text mining.

I. INTRODUCTION

Sentiment Analysis is a common task in the field of Natural Language Processing (NLP). The task aims to detect emotions in text. Previous studies mainly focus on binary (positive and negative) and ternary (positive, negative and neutral) sentiment analysis in monolingual text.

However, in social media such as micro-blogs, emotions are often expressed by text in multiple languages, and the multilingual text is called code-switching text. Detecting emotions from code-switching text is a more complex task than detecting emotions from monolingual text. Moreover, analyzing only two or three kind of emotions is insufficient. People may be happy, sad, angry, afraid or surprised, rather than just have two kind of emotions.

E1 to E5 are five examples to show code-switching text that is mixed with Chinese and English. In these examples, emotions may be expressed by either English or Chinese text, and different emotions may exist together. For instance, happiness is expressed by English text in E1, and the surprise emotion is expressed only by Chinese text in E2. In other cases, both

Chinese and English phrases may contain the same emotion. The fear emotion of E3 is expressed by both Chinese phrase “怎么办” and English phrase “help me”. There are more complicated situations. In E4, a mixed Chinese and English phrase “high 起来” expresses the happiness emotion. Since emotions are hidden inside different languages rather than a single language, it is more difficult to detect them. Moreover, a multi-label sentiment analysis is more complex than a single binary or ternary sentiment classification. Take E5 for example, the code-switching text in E5 contains both sadness and surprise emotions. It contains more than one emotion.

E 1. i'm so happy! 虽然天空还飘着小雨 ~

(I'm so happy, although the day is still under a light rain)

E 2. 啊啊啊啊啊。godiva 的巧克力酒。。发现它出现在 office.

(Ah ah ah ah ah. godiva's chocolate. . Found it appeared in office.)

E 3. help me !!! 那么饿。怎么办。。怎么睡。明天又会赖床了

(help me !!! I'm so hungry. What can I do. . How to sleep. . I will have a lie in tomorrow.)

E 4. 各种亲戚朋友各种好,这真是个幸福快乐的新年! 嘿嘿, high 起来

The associate editor coordinating the review of this manuscript and approving it for publication was Sabah Mohammed^{ID}.

(Hello relatives and friends, this is really a happy new year! Hey, cheer up!)

E 5. 这 bus 的发车时间一会儿早很多, 一会儿晚很多, 到底是闹哪样

(The departure time of this bus is much earlier or later at times. What kind of trouble is it?)

Previous research on sentiment analysis or word classification mainly uses models such as RCNN [1], FastText [2] and CNN [3]. Attention mechanisms [4] are also used to improve the performance of these models. What these researches have in common is the use of word embeddings, including word2vec [5], GloVe [6] or ELMo [7].

Recently, Google has proposed a new sentence-level pre-training model named BERT [8]. BERT is trained on a large amount of unlabeled text by a masked language model for next sentence prediction tasks. It outperforms traditional models in eleven types of NLP tasks, including sentiment analysis.

Most researches on sentiment analysis in code-switching text use methods based on attention mechanisms and LSTM such as BLSTM-MC [9] and BCEL [10]. What these models have in common is the use of word2vec. However, there is little research on using BERT for multi-label sentiment analysis in code-switching text. Therefore, we try to fine-tune BERT for this task in order to get a better performance than most of the models of previous research.

In this paper, we firstly investigate how to adapt and fine-tune BERT for better performance on multi-label sentiment analysis in code-switching text. We compare the performance of different public BERT pre-trained models and different parameters. The languages we mainly focus on are Chinese and English.

For this multi-label multilingual task, there are also many challenges. In Chinese micro-blogs, although people are used to write more Chinese words than English ones to express their feelings, some emotions are expressed in short English sentences. The mixture of different languages causes the difficulties in emotion detection. Another challenge is that different emotions may exist together. The classification model should also learn the relationship between labels. The third challenge is the unbalanced distribution of each type of emotion. Some emotions appear more often than the other ones, so the distribution of samples in different categories is not balanced.

Methods are proposed in our paper to deal with the above challenges. BERT pre-trained models have learned a large amount of semantic information of both languages, so the first challenge can be overcome by the use of BERT. Our experiments focus on the choice of pre-trained models and parameters. Our solution to the second problem is the use of a single BERT model to predict the possibility of all the emotions. To process the unbalanced dataset better, the method of data augmentation, undersampling and ensemble learning are used. We sample the augmented dataset for several times to train multiple classifiers. After sampling, the distribution of samples of emotions is balanced.

Our BERT method for multi-label sentiment analysis in unbalanced code-switching text (BERT-MSAUC) are tested in the code-switching dataset of NLPCC 2018 [11]. Experiments show that our BERT-based approach outperform many previous models.

II. RELATED WORK

Although most research studies are focused on binary or ternary sentiment analysis, there is still some research on multilingual and multi-dimensional sentiment analysis. For example, there are researches on traditional way to solve the task of multilingual text classification [12]. With the development of deep learning, deep learning models including RNN are used for code-switching language modeling [13]. Wang *et al.* propose a term-document bipartite graph [14] to combine both the information of the code-switching text and the sentimental information. Attention mechanisms are also useful in multilingual sentiment analysis. For example, a bilingual attention network [15] is proposed to deal with sentiment analysis in code-switching text, and it achieves a remarkable performance. In 2018, emotion detection in code-switching text is accepted as a common task in an NLP conference named NLPCC 2018. The BCEL model [10] combines the outputs of different binary classifiers such as CNN and RNN to process the multi-label and multilingual sentiment analysis task. This model outperforms many other models in this conference.

Most of the previous deep learning methods are based on word embeddings, but it is difficult to use only one vector to represent polysemous words which have multiple meanings [7]. For example, the word “apple” can refer to a fruit or Apple phones, but it is only mapped to a single vector, which does not distinguish between different semantic meanings. However, with the release of BERT, this problem is solved. BERT pre-trained models learn from a large amount of corpus and generate sentence embeddings rather than word embeddings. As a result, even though the same word may exist different paragraphs, since BERT is able to understand the meaning of sentences, the model may know whether the meanings of these words are the same.

With the release of BERT, many researchers use BERT based methods on sentiment analysis tasks, and the results show that BERT has greatly improved the performance of models in the field of sentiment analysis. Sun *et al.* investigate different BERT fine-tuning methods on the text classification task. They also propose truncation methods and hierarchical methods to process long text that is longer than the maximum sequence length of BERT [16]. Munikar *et al.* propose a solution on fine-tuning BERT for short text fine-grained sentiment analysis [17]. Aspect-Based sentiment analysis has also received great attention these years. For instance, Sun *et al.* [18] transfers the aspect-based sentiment analysis task into a sentence-pair classification problem, and uses BERT-pair to achieve new state-of-the-art results. TD-BERT model [19] is also proposed to deal with the aspect-based sentiment analysis task. Their result shows that coupling

BERT with complex neural networks may not perform well in this task because its performance is sometimes below than the BERT-FC implementation.

However, there is little research on using BERT on code-switching text. In our approach, we fine-tune BERT pre-trained models on our task, and propose effective methods for multilingual and multi-label sentiment analysis.

III. METHODOLOGY

Multi-label sentiment analysis task in code-switching text takes multilingual text as input and outputs sentiment scores of each label. For example, if there are five potential emotions, the output is an array of five scores between 0 and 1 of each emotion. In this section, we first briefly review the standard BERT model, and then explain the BERT(M) and BERT-MSAUC model for this multi-label task.

A. BACKGROUND

BERT (Bidirectional Encoder Representation from Transformers) uses Transformers [4] as language models. The encoding vector of BERT's input is summed by three embeddings. They are token embedding, segment embedding and position embedding. BERT is pre-trained from a large amount of text using two different unsupervised tasks.

1) MASKED WORD PREDICTION

The task randomly covers some input words, and then predicts the word based on context information, which is essentially a cloze problem. During pre-training, 15 percent of the words are randomly masked by the masking symbol [MASK].

2) NEXT SENTENCE PREDICTION

The goal of next sentence prediction task is to learn whether sentence A is followed by sentence B. Note that a document-level corpus rather than a sentence-level corpus is needed.

Different public BERT pre-trained models are trained from corpus of different languages. However, a large number of parameters is needed for the pre-training of BERT, so the time required for training is unimaginable. Fortunately, google provides us with many public pre-trained BERT models, including a BERT-Base multilingual model trained from corpus of 104 languages and a BERT-Base Chinese model. What we need to do is to fine-tune them.

B. BERT(M) FOR MULTI-LABEL SENTIMENT ANALYSIS

We modify the standard BERT model for classification tasks to improve the performance of multi-label sentiment analysis in code-switching text. Unlike binary and ternary sentiment analysis, the output of our task is a five-dimensional vector and different emotions may exist together. Multiple BERT(M) models are used in our BERT-MSAUC model to deal with our task.

1) INPUT LAYER

Since the output of our task is a five-dimensional vector, the simplest method is to add five different classification models to detect each emotion. However, emotions have the possibility of appearing together in our task. The method of using five independent classifications may cause an impossible situation where happiness and sadness exist together. Because the method of using five independent models leads to the lack of relationship between emotions, we use a single BERT model named BERT(M) for multi-label sentiment analysis instead of five ones.

The first layer of BERT(M) is an input layer model. In this layer, the input label is a five-dimensional vector. One way to encode the labels is to convert a vector into a binary number for the purpose of dimensionality reduction. If sadness and surprise emotions exist together in our dataset, the vector is [0, 0, 0, 1, 1] which is encoded as 00011B. However, this method does not consider the association between tags. For example, vectors encoded as 10001B and 10000B both contain the emotion of happiness, but there is no correlation between them since they are encoded as binary numbers. Therefore, in our input layer, we choose to use the method of a five-dimensional vector.

2) CHOICE OF PRE-TRAINED BERT MODELS

Since code-switching text contains different languages, a key problem is the process of different languages. An appropriate pre-trained BERT model is needed in order to get a better performance.

Some available public BERT pre-trained models are trained from both Chinese and English corpus. For example, The BERT-Base multilingual model contains 104 languages, including both Chinese and English. Moreover, the original corpus of the BERT-Base Chinese model is not pure Chinese. It can also recognize many commonly used English words.

In our dataset, most of our code-switching texts contains two languages that are Chinese and English, and there are more Chinese words than English ones. Therefore, the model we choose are a BERT-Base multilingual model and a BERT-Base Chinese model. We fine-tune and compare these two models in our experiments. The details of these models are described in the experiment section.

3) SIGMOID CLASSIFIER

The pre-trained BERT model used in our paper provides us with a 768-dimensional vector as the encoding of the input text vector defined as x . The model connects a fully connected layer W after this vector. The dimension of the fully connected layer is [728, 5]. The output of the model is calculated as follows.

$$z = x \cdot W + b \quad (1)$$

$$y_{prediction} = f_{activation}(z) \quad (2)$$

Activation function is defined as $f_{activation}$, and regular non-linear activation functions include softmax, relu and

sigmoid. Softmax is often used in multi-class tasks, which is the default activation function of traditional BERT. The softmax function is given as below:

$$\sigma(z_j) = \frac{e^{z_j}}{\sum_{k=1}^n e^{z_k}}, \quad j = 1, 2, \dots, n \quad (3)$$

However, softmax function is not suitable for multi-label tasks because the sum of the output probabilities of a softmax activation function is 1. It does not fit the situations where multiple tags may appear at the same time. Another activation function called sigmoid is suitable for multi-label tasks:

$$\sigma(x) = \frac{1}{1 + e^{-x}} \quad (4)$$

The outputs of sigmoid function ranges from 0 to 1, which are independent of each other and present the probability of each label. The sum of the outputs of sigmoid function is not 1. We use sigmoid function as our activation function so that more than one labels have a possibility to get a score of over 0.5. The five-dimensional vector output by the function in our sigmoid layer represents the probability of emotions in five dimensions. The probabilities are finally classified into labels of 0 and 1 with a threshold of 0.5.

C. BERT-MSAUC FOR MULTI-LABEL SENTIMENT ANALYSIS IN UNBALANCED CODE-SWITCHING TEXT

Many previous studies focus on the performance of models on a balanced dataset. However, the reality is that the distribution of samples of different categories is often unbalanced. For example, people may like to share happiness than the other emotions with others in microblogs. Consequently, some emotions appear more frequently than the others, which leads to the reduction of classification performance. In this section, BERT-MSAUC is proposed to deal with the unbalanced code-switching dataset.

BERT-MSAUC is proposed for multi-label sentiment analysis in unbalanced code-switching text. The architecture of BERT-MSAUC is described in Figure 1. The method of data augmentation and undersampling is used to get various balanced datasets to train different BERT(M) models described above. This model combines the output of each classifier to gain the final predictions.

1) DATA AUGMENTATION METHOD BASED ON TRANSLATION

Data augmentation is an effective way in scenarios with few samples or unbalanced categories [21]. When the number of samples does not meet the training requirement, data augmentation methods are often used to increase the size of the dataset, thus improving the performance of a model. Data augmentation methods are also used to obtain more samples for small categories when the distribution of dataset is unbalanced. A more balanced dataset may result in a good generalization ability of a model.

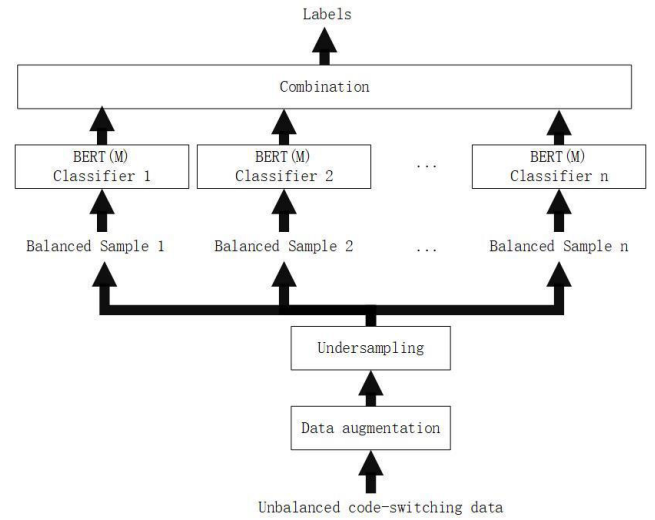


FIGURE 1. The architecture of BERT-MSAUC.

Thanks to the fast development of text translation algorithms and the open source of various advanced translation models such as Google and Baidu api, back translation [22] has become an effective way of data augmentation. For example, if we translate some English text into Chinese, and then translate them back English, we will get various new training data, and the size of the dataset is doubled.

In our approach, we do not modify the test set. Instead, we only expand the training set by adding some translated texts. Since most of our texts are in both Chinese and English, and there are more Chinese words than English ones, we translate the text in our training set into English using Baidu Translation. After data augmentation, the size of the training set is increased and the dataset is more balanced, which helps the model learn more semantic information of both languages in the fine-tuning stage of BERT.

2) MULTIPLE UNDERSAMPLING

Oversampling [23] and undersampling [20] are both common methods to deal with unbalanced dataset, but a simple use of them is not an appropriate solution. An oversampling approach only repeatedly samples a part of data to make the data balanced. Although it seems that the data is balanced, but actually no more data is introduced, which will amplify the impact of noise on the model. Moreover, the use of undersampling may cause the loss of a large amount of data, so it is not an effective approach in a small dataset.

In our approach, a new sampling method is suggested on the basis of the idea of oversampling and undersampling. Our approach is to perform undersampling for n times and train n different classifiers to make full use the unbalanced dataset. We use the method of random undersampling to select the same number of samples from each category. The size of the smallest category is chosen as the number of samples of each category. The benefits of our approach are as follows. From the perspective of each balanced dataset chosen from random

undersampling, samples are not repeated sampled. From the overall point of view, most sentences in our dataset are chosen because we perform undersampling for multiple times. Our sampling approach makes up for the shortcomings of these two sampling methods.

3) ENSEMBLE LEARNING

At present, we have a number of small balanced datasets, and we can only get multiple models with preferences. Since our goal is to learn a stable and good model in all aspects, ensemble learning is introduced in our model to combine the outputs of different classifiers.

Multi-label BERT(M) classifiers for code-switching text are trained with balanced samples. The output of these classifiers is a vector of five dimensions. Each dimension ranges from 0 to 1, representing the probability of each label. Finally, the method of linear combination is selected to combine the prediction results of each model. BERT-MSAUC combines the output of these n labels by calculating the arithmetic mean of each dimension, and the default weight of each classifier is the same.

$$\frac{1}{k} \sum_{i=1}^k w_i p_i(c) \quad (5)$$

IV. EXPERIMENT AND ANALYSIS

A. DATASET AND EVALUATION METRICS

We use a public dataset of NLPCC 2018 shared task 1 [11] for our multilingual and multi-dimensional sentiment analysis method. These texts are from Chinese micro-blogs, and consist of texts of different languages such as Chinese, English, Japanese and many other ones. The training set and validation set of this task are public, but the test dataset is not available. Therefore, we use the original validation set as our test set, and the validation set is not contained in our training set. Examples of code-switching texts of the dataset are shown in E6 and E7.

E 6. 历史 research 搞定之后顿时感觉空虚了很多。。。。(Sadness)

(After the history research was done, I suddenly felt a lot empty. . . ~)

E 7. 天哪! 天哪! 怎么会有这么好看的日剧! 要不要这么好看! 《家政妇三田》一直看到今天早上7点。i love it! (Happiness, Surprise).

(Gosh! Gosh! How could there be such a beautiful Japanese TV show! It is so exciting! I kept watching “Housewife Sanda” until 7 am this morning. i love it!)

In this dataset, five basic emotions are defined, including happiness, sadness, anger, fear and surprise. The training dataset contains 6000 code-switching texts and the test dataset contains 738 texts. Each text may contain zero, one, or multiple emotions. As is shown in Table 1, the distribution of the data is not balanced. Emotions of happiness and sadness appear more frequently than the others. Meanwhile, there are fewer positive examples than negative examples for each type of emotion.

TABLE 1. Data distribution of NLPCC 2018 shared task 1.

Emotion	Training Set	Test Set
Happiness	0.304	0.302
Sadness	0.181	0.165
Fear	0.095	0.115
Anger	0.108	0.117
Surprise	0.109	0.126

Our random undersampling method is to select the same number of samples from each category. The data distribution before and after sampling is shown in Table 2. Since some emotions are more likely to appear with others, there are more samples of happy and sad emotions than the others.

TABLE 2. Data distribution of the training set before and after sampling.

	happiness	sadness	fear	anger	surprise
Before sampling	0.304	0.181	0.095	0.108	0.109
After Sampling	0.253	0.241	0.193	0.201	0.207

B. EXPERIMENT I: FINE-TUNING BERT(M) FOR CODE-SWITCHING TEXT

Our first experiment is to fine-tune BERT(M) for multi-label sentiment analysis in code-switching text. Our experiment focuses on the choice of parameters and BERT pre-trained models. This experiment aims to find the best way to fine-tune BERT(M) for our task, and the fine-tuned BERT(M) will be used in the next experiment.

1) EXPERIMENT SETTINGS

In our experiment, we choose the official Chinese chinese_L-12_H-768_A-12 model and multilingual multi-cased_L-12_H-768_A-12 model as our pre-trained BERT models. We use the final hidden layer of [CLS] for our sentiment analysis to obtain a better representation of the input sequence. Different batch size, learning rate, and epochs are used for fine-tuning BERT. Considering the overfitting problem, we set a dropout rate of 0.1. Most of the important parameters are shown in Table 3, and the parameters of the Chinese and multilingual models are shown in Table 4 and Table 5.

In order to fairly compare the Chinese and multilingual pre-trained models, two BERT pre-trained models with the same parameters are selected in our experiment. These two models both contain 12 layers of transformers and 110M parameters. Their hidden size and number of attention heads are also the same. The only difference is the number of languages. The multilingual model learns from corpus of more languages, leading to a larger vocabulary size.

Since precision, recall, and F1-score are often selected as the evaluation index of sentiment analysis problems, these

TABLE 3. Experiment settings.

Parameter	Value
language model	Chinese, multilingual
dropout rate	0.1
max sequence length	128
word embedding dimension	768
activate function	sigmoid
learning rate	2e-5
epoch	4
batch size	16, 32

TABLE 4. Parameters of Chinese BERT model.

Parameter	Value
number of layers	12
hidden size	768
number of attention heads	12
parameters	110M
vocabulary size	21128
number of languages	2 (Chinese and English)

TABLE 5. Parameters of multilingual BERT model.

Parameter	Value
number of layers	12
hidden size	768
number of attention heads	12
parameters	110M
vocabulary size	119547
number of languages	102

metrics are used to judge the performance of our algorithms in our experiment.

In this multi-label task, the precision, recall and F1-score of each emotion are calculated as follows:

$$precision = \frac{TP}{TP + FP} \quad (6)$$

$$recall = \frac{TP}{TP + FN} \quad (7)$$

$$F1 = \frac{2 \times precision \times recall}{precision + recall} \quad (8)$$

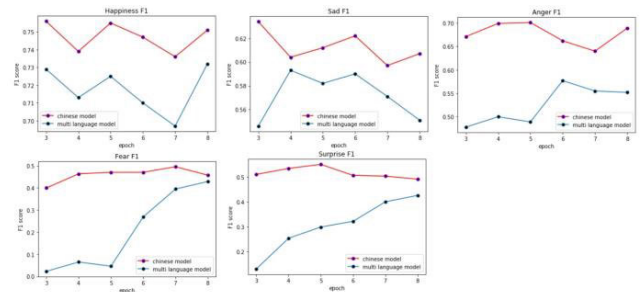
SVM classifier is selected as the baseline of the dataset. The precision, recall and F1-score is calculated for each emotion and the baseline is shown in Table 6.

TABLE 6. SVM baseline.

emotion	precision	recall	F1-score
Happiness	0.610	0.691	0.648
Sadness	0.307	0.650	0.417
Fear	0.299	0.667	0.413
Anger	0.256	0.788	0.386
Surprise	0.142	0.391	0.208

2) COMPARISON BETWEEN DIFFERENT PRE-TRAINED MODELS

The comparison between Chinese model and multilingual model is shown in Figure 2. In this experiment, we choose a batch size of 32 and a learning rate of 2e-5. Result shows that the Chinese model converges faster than the multilingual model. When the number of epochs is 3 and 4, the performance of the Chinese model is already satisfactory, while the multilingual model does not perform well in this task, especially in the emotion of fear and surprise. With the increase of epochs, the performance of the Chinese model is stably good and the performance of the multilingual model is increased, but the performance of the multilingual model is not as good as the Chinese model. Overall, the Chinese pre-trained model, which is trained from corpus of both Chinese and English, outperforms the multi-language pre-trained model in all the five dimensions.

**FIGURE 2.** Comparison between Chinese model and multilingual model.

3) FINE-TUNING BERT(M)

Experiments have been done on different batch size, learning rate and number of epochs on BERT(M). Since the Chinese model has a better performance, we fine-tune the Chinese model using different parameters. It should be pointed out that the Chinese pre-trained model learns from a large amount of Chinese corpus, and some corpus are mixed with Chinese and English. Therefore, it performs well in the code-switching text that contains both Chinese and English.

The result in Table 7 and Table 8 shows that using a batch size of 16 results in a better performance than using a batch size of 32. As for learning rate, a smaller learning rate may cause training to converge more slowly, but it may improve the performance of the model. The learning rate of 2e-5 gets better performance than the other learning rates. Moreover, we selected 3 and 4 as parameters of epochs. According to the experiment, a batch size of 16, a learning rate of 2e-5 and 4 numbers of epochs are the best parameters of BERT(M) for our task.

4) ANALYSIS AND DISCUSSION

In this experiment, we mainly focus on how to fine-tune BERT(M) to achieve a better performance on multi-label sentiment analysis in code-switching text. What we learn from the results are as follows.

TABLE 7. F1-score of BERT(M).

learning rate	epoch	batch size	F1-score
5e-5	3	16	0.558
2e-5	3	16	0.596
5e-5	4	16	0.546
2e-5	4	16	0.602
5e-5	3	32	0.559
2e-5	3	32	0.593
5e-5	4	32	0.535
2e-5	4	32	0.594

TABLE 8. Detailed F1-score of each emotion of BERT(M).

batch size	happiness	sadness	fear	anger	surprise
16	0.773	0.624	0.658	0.452	0.501
32	0.753	0.642	0.667	0.397	0.513

First of all, the training corpus of BERT and the quality of the pre-trained model affects the performance of models in downstream tasks. Although our dataset is made up of texts of more than two languages, most of them is mixed with Chinese and English. Therefore, we can draw the first conclusion that a suitable BERT pre-trained model is not required to conclude all the languages of the texts in the dataset. A pre-trained BERT model that contains most of the languages is enough for this multilingual task. Otherwise, the training speed and the performance may be affected. Therefore, the Chinese model we use in our experiment is suitable for our task since it learns from both Chinese and English corpus. Even though some other languages also exist, our model is able to learn from them in the fine-tuning stage.

In addition, the experiment shows an effective way to fine-tune BERT(M) to achieve a great performance on code-switching text. Compared with the baseline, our model performs better in all the five dimensions. However, the unbalanced distribution of samples of different emotions is not considered in this model, and this is what we focus on in the next experiment.

C. EXPERIMENT II: FINE-TUNING BERT-MSAUC FOR UNBALANCED CODE-SWITCHING TEXT

Our second experiment is to fine-tune BERT-MSAUC for multi-label sentiment analysis in unbalanced code-switching text. Our experiment focuses on the unbalanced distribution of dataset. This experiment aims to check the performance of our BERT-MSAUC model and find out the effect of data augmentation methods and multiple undersampling methods. BERT-MSAUC is compared with BERT(M) and many other popular sentiment classification models.

1) EXPERIMENT SETTINGS

In our experiment, BERT Chinese model is selected as our pre-trained model used in BERT(M). The settings and

parameters of BERT(M) model is the same as the parameters in the above experiment. Arithmetic mean is selected as the combination method of our experiment, and the weight of each sub-model is the same. Nine BERT(M) models are used in order to cover as many contents of the dataset as possible. The detailed experiment settings are listed in the following Table 9.

TABLE 9. Experiment settings.

Parameter	Value
language model	Chinese
dropout rate	0.1
max sequence length	128
word embedding dimension	768
activate function	sigmoid
learning rate	2e-5
epoch	4
batch size	16
combination method	arithmetic mean
number of BERT(M)	9

2) PERFORMANCE OF BERT-MSAUC

According to official evaluation requirements, we calculated the accuracy rate, recall rate, and F1 value of each emotion in the experiment. The official scoring metric of NLPCC shared task 1 is macro-averaged F1-Score. We compared the output of BERT-MSAUC with some of the previous models. The performance of the models is shown in Table 10. Some of the results are from the report in BCEL [10].

TABLE 10. F1-score of each model on the test set.

model	happiness	sadness	fear	anger	surprise
SVM	0.648	0.417	0.413	0.386	0.208
Fasttext	0.600	0.491	0.489	0.350	0.383
RCNN(B)	0.573	0.411	0.454	0.395	0.322
CNN(S)	0.615	0.487	0.606	0.429	0.378
BCEL	0.644	0.527	0.599	0.472	0.391
BERT(M)	0.773	0.624	0.658	0.452	0.501
BERT-MSAUC	0.769	0.614	0.706	0.513	0.497

As is shown in Table 10, SVM is the benchmark of this shared task. Fasttext, RCNN and CNN(S) are binary classification models. Sampling technique is used in CNN(S). Binary classification and ensemble learning are adopted in BCEL, which has a better result than the above models. BERT(M) and BERT-MSAUC are the models proposed in this paper.

BERT(M) and BERT-MSAUC achieve a better performance in all the five dimensions compared with previous models. It is because of the large-scale pre-training and the sentence granularity information learned by BERT. In addition, our BERT-MSAUC model deals with unbalanced datasets better than BERT(M). It has a macro-average F1-score of 0.620, compared with the F1-score

of 0.602 of BERT(M). BERT-MSAUC outperforms the multi-label BERT(M) model in fear and anger emotions, and its performance on the other three emotions is almost the same as the performance of BERT(M). Our approaches on the unbalanced dataset lead to this improvement. Overall, our BERT-MSAUC model achieves the best performance in this experiment.

3) THE EFFECT OF DATA AUGMENTATION AND UNDERSAMPLING

As is shown in Figure 3, with the help of data augmentation, our model achieves a better performance than the original BERT(M) model. The F1-Score of BERT-MSAUC is higher than BERT(M) with data augmentation and BERT(M), especially in the anger and fear emotions. In our dataset, there are fewer fear and anger samples than the others, and the improvement is mainly in these two kinds of emotions. Therefore, the performance shows the effectiveness of our undersampling and combination approaches to deal with the unbalanced dataset.

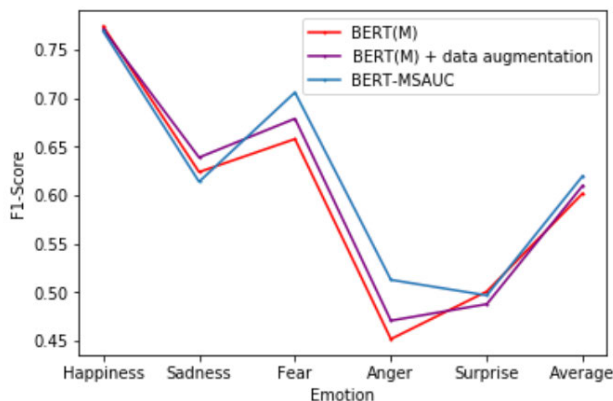


FIGURE 3. The effect of data augmentation and multiple undersampling on F1-score.

Moreover, multiple BERT(M) classifiers are trained with the balanced samples to make full use of all the code-switching text in the dataset. The result in Figure 4 shows the

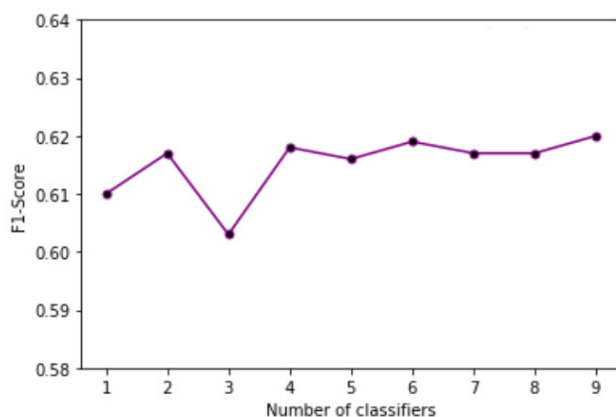


FIGURE 4. Influence of the number of classifiers of BERT-MSAUC on F1-score.

importance of using multiple classifiers. The F1-score in this figure is the macro-average F1-score. When the number of classifiers is fewer than four, the performance of the model is unstable. However, with the increase of number of BERT(M) classifiers, the performance of our model becomes steady. According to the experiment, at least four classifiers are required in our model.

V. CONCLUSION

In this paper, we propose a BERT-based model for multi-label sentiment analysis in unbalanced code-switching text. We use BERT for this task and experiment on different pre-trained models and parameters. Considering the characteristics of the unbalanced dataset, we use a data augmentation method based on translation and a multiple undersampling method to get balanced samples. These samples are trained by several independent BERT classifiers, and ensemble learning is used to combine the outputs of these classifiers.

According to the experiments, we find out that among the public pre-trained models, the BERT-Base Chinese model is more suitable for code-switching text of Chinese micro-blogs than the BERT-Base multilingual model because most of the pre-training corpus is in Chinese or English. Another finding is that 3 epochs, a batch size of 16 and a learning rate of $2e-5$ is enough for our model. Although what we use is a public BERT pre-trained model, the result is still considerable. Moreover, our data augmentation and undersampling method are both helpful in improving the performance of our model in the unbalanced code-switching dataset. Experiment shows the effectiveness of our model on the multilabel and multilingual sentiment analysis task.

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